

Before the Command Was Spoken

Pre-Linguistic Coordination and the Evolutionary Substrate of Law

Ignacio Adrián Lerer

Attorney (UBA), Executive MBA (IAE, Universidad Austral).

Buenos Aires, Argentina.

adrian@lerer.com.ar

ORCID 0009-0007-6378-9749

12 September 2026

Abstract

This paper advances a bounded hypothesis about the deep history of law. It argues that legal ordering did not arise *ex nihilo* with legislation, writing, or fully articulated deontic language. Before humans could formulate general prescriptions, earlier forms of social coordination plausibly relied on partner choice, reconciliation, coalition, retaliation, exclusion, and embodied expectations. These mechanisms should not be redescribed as law without qualification. They are better understood as an evolutionary substrate from which proto-normative practices and, much later, recognizably legal institutions could develop. The paper distinguishes six layers: pre-linguistic coordination, proto-normativity, embodied subjective selection, deontic linguistic scaling, institutionalization, and recursive institutional feedback. Language is treated as a candidate amplification technology that increased abstraction, portability, contestability, and third-party enforcement rather than as the proven cause or origin of normativity. Extended Phenotype Theory and Evolutionary Game Theory enter only after their units and mechanisms are specified: institutions may alter the environment in which normative variants are transmitted, while game-theoretic models may identify conditions of stability or invasion. The argument is theoretical and abductive. It does not establish that full law existed before language, that any legal norm is a meme, or that historical persistence demonstrates evolutionary stability. Its contribution is a testable architecture for investigating continuity and discontinuity between non-linguistic coordination, human norm psychology, and legal institutions.

Keywords: legal evolution; proto-normativity; pre-linguistic coordination; deontic language; cultural evolution; extended phenotype; evolutionary game theory; subjective selection

1. Introduction

Legal theory usually begins after several difficult problems have already been solved. Its canonical objects are commands, rules, reasons, precedents, offices, procedures, and texts. Even customary law is commonly described through concepts that presuppose public recognition and linguistic articulation. This starting point is appropriate for doctrinal analysis, but it may be too late for an evolutionary account of legal ordering.

The historical question is not whether a statute could exist without words. It could not. Nor is it whether contemporary legal validity can be reduced to primate behavior. It cannot. The narrower question is whether some of the coordination problems and behavioral mechanisms later organized by law predated the capacity to formulate explicit prescriptions.

The hypothesis developed here is that they did. Partner selection, retaliation against direct harm, reconciliation after conflict, coalition formation, exclusion, and repeated cooperative routines can operate without a general rule stated in language. Evidence from nonhuman primates does not establish human normativity, but it makes a more modest proposition plausible: evolution had assembled components of

social regulation before humans developed legal institutions. Human societies may later have reorganized those components through shared intentionality, ritual, language, authority, and material infrastructure.

This paper revises a stronger earlier formulation. It does not claim to demonstrate that law itself preceded language. It distinguishes a pre-linguistic substrate from proto-normativity and both from positive law. It also withdraws the inference that the absence of a particular archaeological marker proves egalitarian enforcement, and it does not use a single group-size estimate as a causal threshold. These corrections narrow the thesis but improve its explanatory value.

The revised argument proceeds in six layers. First, pre-linguistic social mechanisms created patterned coordination. Second, some human groups may have transformed such patterns into proto-normative expectations sustained by approval, disapproval, sanction, and repair. Third, people selectively retained embodied procedures that appeared to solve recurring problems. Fourth, deontic language increased the reach and revisability of normative expectations. Fifth, roles, records, procedures, and authorities stabilized complex legal packages. Sixth, those institutions modified the environment in which later norms were produced and selected.

The contribution is therefore not an origin story stated as fact. It is a comparative research architecture. Its value depends on whether it generates distinctions and predictions that rival accounts do not.

2. Four Concepts That Must Not Be Collapsed

2.1 Coordination

Coordination occurs when behavior becomes mutually adjusted. It may arise through repetition, signaling, dominance, mutual benefit, kinship, or environmental constraint. Coordination alone does not imply obligation. Two animals can converge on a stable pattern without either treating deviation as a violation.

2.2 Proto-normativity

Proto-normativity refers here to socially sustained expectations that organize conduct and elicit patterned responses to deviation, without presupposing explicit general rules or a legal system. The term names a research category, not a discovered prehistoric institution. It becomes empirically useful only if researchers can distinguish expectation, sanction, repair, and transmission from habit or dyadic bargaining.

2.3 Normativity

Normativity introduces a claim about what an actor should do, not merely what actors usually do. Philosophical accounts disagree about whether language, shared intentionality, second-personal authority, reasons, or communal recognition is necessary. The present argument does not resolve that dispute. It asks which observable capacities could have supported later normative practices.

2.4 Law

Law is used in its recognizably institutional sense: normative ordering involving differentiated roles, procedures, public criteria, authoritative settlement, remedies, or organized coercion. Some legal systems are oral, but they are not therefore pre-linguistic. Calling every stable social regularity “law” would erase the phenomenon the theory aims to explain.

These distinctions create a continuity thesis without assuming identity. Pre-linguistic coordination may contribute to proto-normativity; proto-normativity may contribute to law; neither transition is automatic.

3. The Pre-Linguistic Substrate

3.1 Partner choice and exclusion

Chimpanzees recruit collaborators selectively and adjust cooperation to tolerance and prior success (Melis, Hare, & Tomasello, 2006a, 2006b). Such findings support the existence of partner-control mechanisms. They do not show that chimpanzees possess a rule that defectors ought to be excluded. The same behavior can result from learned payoff expectations without norm representation.

For the legal-origin hypothesis, the relevance is functional and limited. Partner choice changes the cost of noncooperation. A repeatedly excluded individual receives fewer opportunities. This mechanism could stabilize cooperative patterns before explicit third-party rules, especially in small networks where encounters recur and reputational information travels through observation.

3.2 Retaliation, policing, and third-party enforcement

Experimental work reviewed in the earlier version found stronger evidence for retaliation to direct harm than for disinterested third-party punishment in chimpanzees (Riedl et al., 2012). The distinction matters. Second-party retaliation can regulate conduct without a generalized norm. Third-party enforcement suggests a broader capacity to treat another's violation as a matter for collective response, but even there researchers must distinguish impartial enforcement from status maintenance, conflict suppression, kin protection, or anticipated benefit.

The absence of third-party punishment in a bounded experiment cannot date its emergence in human evolution. Nor can it prove that prescriptive language caused it. It identifies a gap to be explained: human communities developed forms of third-party mediation and enforcement that appear much more elaborated than those observed in other primates.

3.3 Reconciliation and relationship repair

Reconciliation provides a second substrate. Post-conflict affiliative behavior can restore valuable relationships without a tribunal, a stated rule, or a concept of compensation. Its importance becomes clearer in light of research on small-scale justice institutions. Fitouchi and Singh (2023) argue that some punitive institutions are better understood as devices for restoring reciprocal cooperation than as mechanisms for maximizing deterrence.

The continuity is not that primate reconciliation already contains law. It is that relationship repair presents a recurrent social problem and a family of responses upon which human cultural selection can work. Later institutions can add proportionality, authorized third parties, compensation, ritual closure, and public memory.

3.4 Coalitions and power

Coalitions can suppress aggression, reverse dominance, or reinforce hierarchy. Their ambiguity warns against treating every collective sanction as a norm. A coalition may stabilize cooperation for its members while imposing severe costs on outsiders. Any evolutionary account of law must therefore incorporate power from the beginning. The question is never only which rule appears useful, but useful to whom, selected by whom, and imposed on whom.

4. From Behavioral Pattern to Proto-Norm

Repeated behavior becomes theoretically more interesting when four elements converge: a shared expectation, a response to deviation, transmission beyond a single dyad, and some degree of generalization. None alone is sufficient. Together, they provide an operational starting point for investigating proto-normativity.

Embodied practices may carry expectations without expressing a complete proposition. Gesture, sequence, timing, spatial arrangement, withdrawal, ridicule, and ritual can make conformity visible and deviation costly. Yet ethnographic cases cannot be projected unchanged into the Paleolithic. Contemporary small-scale societies have language, histories, exchange networks, and contact. They can illuminate mechanisms, not provide direct windows into prehistoric mental states.

This qualification changes how evidence should be assembled. Primatology identifies possible ancestral components. Developmental psychology examines how children acquire and enforce expectations. Ethnography shows how nonwritten institutions transmit obligations. Archaeology constrains material possibilities and temporal sequences. No single stream proves the transition. The argument is abductive and must survive explanations based on ecology, coercion, individual learning, shared ancestry, and diffusion.

5. Subjective Selection Before and After Words

Singh (2026) proposes that complex cultural traditions often emerge through subjective selection: people produce and retain variants they evaluate as useful for achieving goals. The framework is particularly relevant to legal origins because social groups repeatedly confront injury, betrayal, uncertainty, allocation, and conflict.

Before explicit legal language, subjective selection could operate on embodied sequences. A group might retain a recurring exchange, reconciliation ritual, avoidance practice, or coalition response because participants perceived that it calmed conflict, restored cooperation, satisfied retributive impulses, or protected valued partners. Conscious theory was unnecessary. Evaluation could be habitual, affective, vicarious, or based on noisy feedback.

This account does not imply that the selected procedure objectively worked. Nor does it imply equal influence. Powerful actors can restrict which variants are proposed, amplify their preferred evaluation, and enforce arrangements that serve them. Institutions may persist because they appear effective to decision makers while burdened groups experience them differently.

Subjective selection supplies a plausible proximate mechanism between recurrent social problems and recurrent normative forms. It does not by itself explain inheritance fidelity, strategic stability, or institutional feedback. Those are the points at which EPT and EGT may add value.

6. The Deontic Threshold as Amplification

The earlier version described prescriptive language as a high-fidelity transmission technology. That formulation remains a hypothesis, but it requires sharper boundaries.

Language can plausibly transform normative ordering in at least four ways. First, it allows a community to detach an expectation from the immediate scene. Second, it permits generalization across persons and events. Third, it makes exceptions and reasons contestable. Fourth, it enables absent third parties to learn, interpret, and enforce a rule.

These capacities support scale, but the causal sequence remains uncertain. Norm psychology may have facilitated language; language may have amplified norm psychology; both may have coevolved with teaching, ritual, reputation, and collective action. A linear story in which proto-norms caused language would exceed the evidence.

The revised hypothesis is therefore reciprocal and time-indexed:

coordination_t -> shared expectations_t -> communicative capacity_{t+1} -> deontic abstraction_{t+1} -> broader enforcement_{t+2}

Feedback runs in the other direction because linguistic rules reorganize attention, classification, memory, and sanction. Future research should compare models rather than assume the arrow.

7. From Norms to Institutions

Legal institutions are complex packages. They combine roles, procedures, remedies, records, symbols, and enforcement capacities. Singh treats justice institutions as candidate super-attractors because societies repeatedly develop structured responses to transgression that impose costs, transfer benefits, use proportionality, and follow institutionalized procedures.

This is compatible with the present account if “justice” is treated as a family rather than a single universal form. Different societies may converge functionally on closure, repair, deterrence, vengeance, status restoration, or public reassurance while diverging sharply in doctrine and morphology.

The transition to law occurs when normative practices acquire a durable division of labor and a public infrastructure of settlement. Oral institutions can satisfy this condition. Writing greatly expands persistence and reach but is not conceptually necessary.

8. Extended Phenotypes and Institutional Feedback

Extended Phenotype Theory provides a disciplined question: does a replicating pattern produce effects outside its immediate carrier that systematically alter its prospects of transmission? Applied to law, the language must remain conditional. A courthouse or law school is not automatically the phenotype of a meme merely because it transmits law.

A defensible EPT analysis requires a candidate informational pattern, identifiable carriers, reproduction or reconstruction events, variation, differential persistence, and an external effect that changes subsequent transmission. Institutional archives, professional credentials, precedent networks, ritualized hearings, and enforcement routines are plausible mechanisms because they shape which interpretations remain available and authoritative.

The institutional feedback loop is:

normative variant_t -> institutional arrangement_t -> altered information and incentives_{t+1} -> differential reproduction of variants_{t+1}

The institution does more than preserve a rule. It can create categories, interests, expertise, and expectations that make certain future rules easier to formulate and others harder to imagine. This is the strongest sense in which legal ordering is recursive.

9. Evolutionary Games Without Metaphorical Stability

Evolutionary Game Theory can formalize competition among strategies, but historical survival does not establish an Evolutionarily Stable Strategy. A valid application must specify players or populations,

available strategies, interaction structure, payoffs, updating rule, enforcement probability, and mutation or entry.

Consider a simplified restoration game. After a transgression, the victim can retaliate, withdraw, or accept repair. The transgressor can deny, compensate, submit to mediation, or escalate. Third parties can remain passive, mediate, or punish. A procedure becomes stable only under stated conditions, such as sufficiently valuable future interaction, credible enforcement, acceptable repair costs, and limits on opportunistic claims.

The model can then compare hypotheses. A restoration account predicts mediation when future interdependence is high. A deterrence account predicts sanctions calibrated to expected prevention. A dominance account predicts procedures favoring actors who control enforcement. A signaling account predicts visible punishment even when private deterrence is weak.

This method replaces retrospective labeling with discriminating structure. If the procedure persists only through coercive exclusion of alternatives, that may be institutional lock-in, not an ESS in the canonical sense.

10. Predictions and Research Design

The revised theory yields seven predictions.

First, low-scale, high-interdependence settings should display stable repair sequences before or without abstract legal formulation. Evidence of mere repeated exchange without shared expectations would count against proto-normativity.

Second, stronger deontic resources should correlate with portability, generalization, explicit exceptions, and third-party enforceability rather than with the first appearance of all social regulation.

Third, noisy or delayed feedback should increase the persistence of procedures that appear effective but do not improve objective outcomes.

Fourth, normative arrangements that create archives, specialists, and authorized interpretive channels should persist longer than initially similar arrangements lacking such infrastructure, after controlling for age, coercion, and prior power.

Fifth, institutions should change how participants formulate their goals. A successful theory should detect preference formation, not assume fixed preferences.

Sixth, independent societies should converge more strongly on functions such as repair, proportionality, and conflict closure than on identical doctrinal forms.

Seventh, institutional outcomes should disproportionately reflect the evaluations of actors controlling communication and enforcement.

A comparative program would code cases across three domains: nonhuman social regulation, small-scale human dispute settlement, and historical legal institutions. The design must control for linguistic relatedness, geographic diffusion, colonial imposition, ecology, subsistence, network density, and political hierarchy. The result would not prove a single origin, but it could reject overly simple continuity or discontinuity accounts.

11. Objections

11.1 Normativity requires language

If normativity necessarily requires propositional representation, pre-linguistic proto-normativity may be a category error. The response is not to redefine law downward. It is to test whether socially generalized expectations and third-party responses can exist without explicit deontic language. If they cannot, the continuity thesis must stop at coordination.

11.2 Functional continuity is not genealogical continuity

Primate reconciliation and human mediation may solve similar problems without one being ancestral to the other. This objection is correct. Functional analogy establishes a candidate mechanism, not descent. Phylogenetic, developmental, and archaeological evidence must carry the genealogical claim.

11.3 Institutions reflect power, not shared goals

Power is not an external correction to the model. It affects which goals matter, which variants enter circulation, and whose evaluations become authoritative. A legal institution may be stable and recurrent because it serves rulers. Subjective selection must therefore be weighted by influence and enforcement capacity.

11.4 Memes and phenotypes anthropomorphize institutions

Norms do not literally want to reproduce. Replicator language is permissible only as shorthand for observable transmission dynamics. When no unit, copying process, variation, or differential persistence can be identified, EPT should be withheld.

11.5 Evolution explains existence but not legitimacy

Correct. An origin or persistence account cannot establish legal validity, justice, or moral authority. A norm may be stable because it coordinates domination. Normative evaluation requires separate legal and moral premises.

12. Conclusion

Law did not emerge from nothing when the first command was spoken. But it does not follow that law, fully formed, existed before words. The defensible claim lies between those extremes.

Evolution supplied mechanisms of coordination and relationship management. Human social cognition may have transformed some of them into shared expectations. Subjective selection could retain practices perceived as useful for recurrent conflicts. Language then made obligations more abstract, portable, contestable, and enforceable beyond the immediate encounter. Institutions gave those obligations roles, procedures, memory, and force. Once established, institutions altered the environment in which later norms competed.

This layered account connects pre-linguistic behavior to legal ordering without erasing the discontinuities that make law distinctive. It also turns a speculative origin story into a research program. The theory should survive only if its layers can be distinguished empirically, its causal arrows outperform rivals, and its evolutionary vocabulary does explanatory work that ordinary institutional analysis cannot.

References

- Boehm, C. (1999). *Hierarchy in the Forest: The Evolution of Egalitarian Behavior*. Harvard University Press.
- Boyd, R., & Richerson, P. J. (1985). *Culture and the Evolutionary Process*. University of Chicago Press.
- Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.
- Dennett, D. C. (2017). *From Bacteria to Bach and Back: The Evolution of Minds*. W. W. Norton.
- Fitouchi, L., & Singh, M. (2023). Punitive justice serves to restore reciprocal cooperation in three small-scale societies. *Evolution and Human Behavior*, 44, 502-514. <https://doi.org/10.1016/j.evolhumbehav.2023.03.001>
- Maynard Smith, J. (1982). *Evolution and the Theory of Games*. Cambridge University Press.
- Melis, A. P., Hare, B., & Tomasello, M. (2006a). Chimpanzees recruit the best collaborators. *Science*, 311, 1297-1300.
- Melis, A. P., Hare, B., & Tomasello, M. (2006b). Engineering cooperation in chimpanzees: Tolerance constraints on cooperation. *Animal Behaviour*, 72, 275-286.
- Richerson, P. J., & Boyd, R. (2005). *Not by Genes Alone: How Culture Transformed Human Evolution*. University of Chicago Press.
- Riedl, K., Jensen, K., Call, J., & Tomasello, M. (2012). No third-party punishment in chimpanzees. *Proceedings of the National Academy of Sciences*, 109, 14824-14829.
- Singh, M. (2026). Subjective selection, super-attractors, and the origins of the cultural manifold. *Behavioral and Brain Sciences*, 49, e285. <https://doi.org/10.1017/S0140525X25100617>
- Tomasello, M. (2009). *Why We Cooperate*. MIT Press.